

HEAPLINE · DESIGN DECISIONS REGISTER

HeapLine design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Register opened; design decisions made under Amish's pre-approvals of 2026-10-03; four requirement decisions proposed, awaiting Amish

HeapLine design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in `docs/REVIEW.md`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list decisions.

Safety: Several decisions below set HeapLine's safety case (lookout and stop rule, capstan overload limit, holding, anchors, what the capstan may pull, blunt probes, no lithium cells). Each took the conservative option; the evidence that would relax it is in HPL-DDR-001, Table 1.

■ Open decisions

The four requirements that are at risk or not met on paper are Amish's to decide (Amish, 2026-10-03: "A simple statement doesn't add value - ensure you are identifying a state and posing it as a clear recommendation for me to decide on."). The state and the effect of each option are set out in `docs/REVIEW.md`, TRL 3, Decisions for Amish.

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	R2 probe depth, at risk in stiffer debris	A: 18 mm tip on all six probes. B: three probes with 22 mm tips and three with 18 mm tips. C: keep the 22 mm tip and measure debris resistance on a test heap first	A, confirmed on a test heap at TRL 4. Proposed, awaiting Amish	Probe tip turning (section 3.3)	HPL-CAL-001, B; REVIEW.md
2	R5 casualty transport, at risk with two haulers on debris	A: drill card rule of four haulers on debris, two only along the boards. B: add a 10 m haul line with shoulder loops. C: both	A. Proposed, awaiting Amish	Drill card wording; line 24 if B	HPL-CAL-001, D; REVIEW.md
3	R8 portable, at risk for four people in one trip	A: two waves, the search kit first and the capstan set by a second group. B: lighter boards. C: a two-wheeled handcart	A. Proposed, awaiting Amish	Drill card and carry bags; board battens if B	HPL-CAL-001, F; REVIEW.md
4	R9 cost, not met	A: one capstan set shared by neighbouring sites. B: a reused crate or bought job box and PPE from the cooperative's stock. C: both	C. Proposed, awaiting Amish	Box and capstan sets per site	HPL-CAL-001, G; REVIEW.md

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Release load of the shear pins from the bar actually bought	The 6 mm pin is sized on 258 MPa shear; the lower end of its band must stay above 5 kN (R4) and the upper end below 8 kN (R10)	HPL-CAL-001, A4
2	Breaking strength of the rope and its splice	Factor 4.5 at 8 kN assumes 40 kN and 90 % at the splice	HPL-CAL-001, A5
3	Working load limits and pin sizes of the slings and shackles bought	The anchor eye's 22 mm hole is drilled for a 19 mm shackle pin	HPL-DWG-104
4	Outside sizes of the tubes bought (60.3, 76.1, 88.9, 114.3)	The acetal bushes are turned to the bores	HPL-DWG-110
5	Plywood thickness and the C16 grade of the battens	Board sinkage and void bridging assume 15 mm plywood on 45 x 70 C16	HPL-CAL-001, C
6	Friction of the HDPE sheet bought on boards and mixed waste	The haul forces of R5 assume 0.3 and 0.5	HPL-CAL-001, D
7	Sizes of the carry bags	The packing layout in HPL-DWG-002 uses their outer sizes	HPL-DWG-002

■ Value engineering

Value-engineering target: USD 1,800 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 1,599.00 (USD 201.00 under the target). Main cost drivers and savings worth trying:

- The capstan set (frame, drum, ratchets, head, pawls, bars, bushes, roller, keeper, pins, stakes, rope, slings and clamp) is USD 449, the largest block.
- The site box with its hardware is USD 253; it only has to keep the kit dry and locked.
- PPE and lighting together are USD 263.
- Bought lifting gear (slings and shackles, USD 66) and the rope (USD 72) stay at rated grades because they carry the safety case.
- Savings worth trying: a reused crate or steel job box for the site box; buying PPE in bulk across cooperatives; profile cutting the ratchets and pawls for several kits at once; a local turner for the probe tips.

■ Decisions made

Date	Decision	Decided by	Record
2026-10-03	Vertical friction capstan pumped through a quarter turn with a drive ratchet, in place of the SiltHaul block or a walk-round capstan	Amish: "I pre-approve the batch runs along with any recommendations you come up with." and "Proceed with the remaining 15 scaffolds"	HPL-DDR-001, items 1, 14
2026-10-03	6 mm S275 shear pin limits the rope to about 6.4 kN; every part sized on 8 kN; R10 added	Amish, same pre-approvals	HPL-DDR-001, item 2
2026-10-03	Two holding pawls on a ring at the drum's foot; tail made fast on a cleat when hauling stops; R11 added	Amish, same pre-approvals	HPL-DDR-001, item 3
2026-10-03	Capstan anchored only by a sling to a sound tree, vehicle or column; stakes only against turning	Amish, same pre-approvals	HPL-DDR-001, item 4
2026-10-03	14 mm polyester double braid, 40 kN	Amish, same pre-approvals	HPL-DDR-001, item 5
2026-10-03	The capstan pulls only sheeting, timber and debris sideways; never lifts, never pulls on a person	Amish, same pre-approvals	HPL-DDR-001, item 6
2026-10-03	Sectional steel probes pushed by hand with a blunt tip; never driven	Amish, same pre-approvals	HPL-DDR-001, item 7
2026-10-03	Plywood crawl boards on battens, linked end to end	Amish, same pre-approvals	HPL-DDR-001, item 8
2026-10-03	Made HDPE sheet stretcher	Amish, same pre-approvals	HPL-DDR-001, item 9
2026-10-03	AA alkaline lamps; no lithium cells in the box	Amish, same pre-approvals	HPL-DDR-001, item 10
2026-10-03	No gas detector in the base kit; no flames; stop on any sign of gas	Amish, same pre-approvals	HPL-DDR-001, item 11
2026-10-03	Six-signal stop rule; lookout posted before anyone steps onto debris; safe zone to the side	Amish, same pre-approvals	HPL-DDR-001, item 12
2026-10-03	First co-design candidate to approach: Uganda Red Cross Society in Kampala with the Kiteezi waste pickers (not agreed)	Amish, same pre-approvals	HPL-DDR-001, item 13
2026-10-03	CalRig as the first candidate rig for capstan proof loads and pin release tests	Amish, same pre-approvals	HPL-DDR-001, item 14
2026-10-03	Combination padlock on the site box; monthly inspection log	Amish, same pre-approvals	HPL-DDR-001, item 15
2026-10-03	budget_usd kept at 1,800 as a value-engineering target	Amish: "I also accept any cost overruns or variations from the assumed scope cost."	HPL-DDR-001, item 16

Date	Decision	Decided by	Record
2026-10-03	Design for construction: the fifteen changes of HPL-DDR-002	Amish: "I pre-approve the batch runs along with any recommendations you come up with." and "Proceed with the remaining 15 scaffolds"	HPL-DDR-002
2026-10-03	Appearance model additions for renders: rope turns on the drum, drill card on the lid, a mannequin beside the capstan	Amish, same pre-approvals	docs/ REVIEW.md, TRL 3